

Mohammed Aljomaily

Computer Engineering Student | Embedded Systems, PCB Design & RF

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Summary

Computer Engineering student with hands-on hardware R&D across PCB design and home fabrication, embedded firmware, and RF systems. Built and hand-etched a dual-transceiver ESP32 RF board in KiCad, a 5-inch FPV quadcopter, and RC aircraft. Fluent in C/C++ on ESP32 and ESP8266, with 14 months of professional engineering experience.

Hardware & Embedded Projects

- **5-Inch FPV Quadcopter:** Assembled, configured and tuned a 5-inch quadcopter (Mario 5 frame, Mamba MK4 F722 flight controller, F40 V2 motors, DJI O4 Lite video, 6S). Configured Betaflight and an ExpressLRS 2.4 GHz control link, and wrote its preflight checklist.
- **Dual-Transceiver ESP32 RF Board:** Designed and hand-fabricated a 2-layer KiCad board for educational study of 2.4 GHz protocols, carrying two nRF24L01+PA+LNA transceivers on independent SPI buses. Authored custom footprints and reworked via geometry for home etching after finding the original layout impractical to produce.
- **Flight Telemetry & Attitude Indicator:** Built an ESP32 system fusing an MS5611 barometer and MPU6050 IMU through a complementary filter, with a Python and Qt QML ground station rendering a live artificial horizon.
- **Discrete Transistor Logic Gates:** Designed and fabricated a resistor-transistor logic (RTL) board with AND, NOT, OR and NAND gates from discrete PN2222A transistors. Fully routed 2-layer board; first revision under debug.
- **RC Aircraft & Radio Link:** Designed and built RC aircraft from first principles: airframe layout, carbon spar integration, servo geometry, and 3D-printed parts. Wrote 4-channel ESP-NOW firmware with a signal-loss failsafe.
- **Router Firmware Reverse Engineering:** Located the UART header on a Xiaomi router and captured boot logs over it, then extracted the firmware image from flash after finding the console locked.

Experience

Software Engineering Intern – Emerging Technologies Department

Applab.qa, Doha, Qatar (client: Al Jazeera Media Network)

March 2025 – April 2026

- Primary author of the backend for Al Jazeera's AI video editing and multilingual translation platform, delivering the API service, the asynchronous job worker, and the projects API.
- Architected an event-driven pipeline across Azure Container Apps and Storage Queues, with a decoupled worker consuming queued work against Cosmos DB state.
- Built failure handling for a long-running distributed pipeline: poison-message routing, bounded retry with backoff, and per-action state logging.
- Integrated Azure Cognitive Services Speech and Azure OpenAI, supporting transcription and translation across 14 locales.

Technical Skills

Hardware: PCB design & layout, schematic capture, custom footprints, home fabrication (toner transfer, etching, drilling), through-hole assembly, FDM 3D printing; KiCad 9/10, FreeCAD, BambuStudio

Embedded: ESP32, ESP8266, Arduino framework, SPI, I²C, UART, PWM/servo, ADC

RF & Protocols: nRF24L01+PA+LNA, ESP-NOW, ExpressLRS, Bluetooth/BLE, 2.4 GHz ISM, IR

Languages: C/C++ (embedded), Python, Go (Golang), C (Linux)

Software: Azure, Docker, Linux (Debian/Arch), Git

Education & Certifications

B.Sc. Computer Engineering – Istanbul Aydin University, Istanbul, Turkey

2025 – Present

IGCSE & A-Levels: Physics, Mathematics, Computer Science

PCEP: Entry-Level Python Programmer (2026) | Google Go Specialization, UC Irvine (Coursera)